Research for the ProveIt project

Removing Redundancy from Diophantine Consistency Equations

Unweighted Pell radicals, stronger transport bounds, shared integer guards, and exact norm identities

Prepared for Vladimir Reshetnikov
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Abstract

We construct and verify several improvements to explicit polynomial equations whose unsolvability is equivalent to syntactic consistency of ZFC. A direct quartic over the unrestricted integers is reduced from 604 to 589 unknowns and from 9,676 to 8,156 monomials, at unchanged coefficient height. A square-gap theorem removes an asymmetric radical weight, and an auxiliary Pell index bound eliminates a separate transport majorant. Together they give a ten-positive-variable equation with 96 arithmetic operations and exact degree 360, and a direct-base companion with 102 operations and exact degree 332. Exact norm identities give pointwise-identical integer evaluators with 126 operations in thirteen variables, and 138 or 137 in twelve variables.

Complete equations, fixed coefficient payloads, source pins, builders and exact checkers accompany the report. All comparisons use one frozen repository snapshot. These results improve specific complexity measures; they do not establish global minimality or reduce the repository's best variable count. The arithmetic-to-refutation calibration is inherited. No new end-to-end Lean, Rocq or object-language ZFC derivation is claimed.

AI-assisted research. Unbounded mathematical arguments are stated separately from exact algebraic and finite-specification checks.

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1 Results, domains and the meaning of simplicity

The objective is a fixed polynomial P with integer coefficients and a precisely stated domain D for which ordinary mathematics establishes

$$\mathrm{ZFC} \vdash [(\forall \boldsymbol{x} \in D^n)\, P(\boldsymbol{x}) \neq 0] \iff \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

Here Con(ZFC) means that the specified syntactic ZFC calculus has no finite derivation of falsity. We neither assume nor prove Con(ZFC). A proof of the biconditional is different from a proof of its left side.

There is no established scalar measure of the *simplest* Diophantine equation. Variable count, degree, arithmetic operations, coefficient height and expanded support measure different costs. We preserve the domain and compare those quantities separately. Ten positive variables are not ten unrestricted integer variables.

Domain and construction	Variables	Old degree	New degree	Operations
Positive, scaled base	10	368	**360**	99 → **96**
Positive, direct base	10	340	**332**	105 → **102**
Integers, quadratic trace	13	536	536	135 → **126**
Integers, rational carrier	12	1,200	1,200	145 → **138**
Integers, integral carrier	12	1,456	1,456	141 → **137**

Table 1: Exact circuit comparisons. Each live binary addition, subtraction or multiplication costs one operation, including multiplication by a constant. Constant leaves and register reuse are free.

The direct quartic gives a different improvement:

	parent	new
unrestricted integer variables	604	589
quadratic residuals	372	371
total degree	4	4
collected monomials	9,676	8,156
maximum absolute coefficient	2,445,790,974,704	2,445,790,974,704

(2)

The saving is fifteen actual variables after projection, and 1,520 monomials, approximately 15.71% of the parent's support. All 8,156 coefficients are supplied explicitly.

The comparison repository is `VladimirReshetnikov/ProveIt`. The frozen commit is

`84e16e43092614bae262ca59651cb9d77e31a255`.

Its active research is newer than its placed manuscript collection. The 99/368 and 105/340 positive endpoints, thirteen-integer trace, two twelve-integer carriers, and 75-predicate quartic family are the actual parents used here [1, 2, 3, 4, 5].

1.1 Different kinds of equivalence

The integer norm rewrites are *identities of polynomials*: values and zero sets agree at every assignment, including degenerate ones. Removing the radical weight and simplifying the transport carrier changes the positive polynomial; we prove equivalence of *positive zero existence*, not equality of zero sets at

unchanged inputs. The quartic transformations change certificate variables and reconstruct every old sign condition before invoking the source interpretation.

The contributions are new improvements relative to the inspected snapshot. We do not claim worldwide priority for a generic norm identity or an elementary sign lemma. The finite repository audit is not an exhaustive historical search.

1.2 Inherited consistency calibration

The positive equations and the small integer norm equations inherit the repository's ordinary native-compiler/K215/refutation calibration. The direct quartic uses its separate C222 proof-certificate route and does not depend on K215 or the Jones nine-variable compression. The two routes are not identified.

The new finite algebraic transformations and ordinary witness-existence arguments are formalizable in ZFC. Composing them with their pinned source theorems gives (1). Exact checkers do not replace the inherited unbounded calibration proof, and this report does not supply the complete encoded derivation in the project's object-language ZFC calculus.

Small-variable universal constructions are separate substantial results; Sun's ten-integer theorem illustrates the distinction between reducing variable count and simplifying a particular evaluator [7]. The present packet does not improve that general bound or the repository's nine-positive-variable endpoint.

2 Exact four-root norm identities and smaller integer evaluators

For formal radicands a, b, c, d, define

$$F_{16}(x; a, b, c, d) = \prod_{\epsilon \in \{-1,1\}^4} (x - \epsilon_1\sqrt{a} - \epsilon_2\sqrt{b} - \epsilon_3\sqrt{c} - \epsilon_4\sqrt{d}). \tag{3}$$

Changing the sign of a root permutes the factors. The product is therefore an integer polynomial, even in x. Write $N_4(z; a, b, c, d) = F_{16}(x; a, b, c, d)$ when $z = x^2$.

2.1 A symmetric identity

Compute

$$\begin{gathered} p = a + b, \quad q = c + d, \qquad u = ab, \qquad v = cd, \\ e_1 = p + q, \quad e_2 = u + v + pq, \quad e_3 = uq + vp, \quad e_4 = uv, \\ s = z - e_1, \qquad t = s^2 - 4e_2, \qquad C = t^2 + 64(e_4 - ze_3), \qquad L = t - 4zs. \end{gathered} \tag{4}$$

Theorem 2.1. *The following is an identity of integer polynomials:*

$$\boxed{N_4(z; a, b, c, d) = C^2 - 256e_4L^2.} \tag{5}$$

The displayed schedule uses 29 binary operations when z is supplied.

Proof. For algebraically independent r_1, r_2, r_3, r_4, put $a_i = r_i^2$ and $x = r_1 + r_2 + r_3 + r_4$. Let p_2, p_3, p_4 be the elementary symmetric expressions in the r_i. Direct expansion gives

$$2p_2 = x^2 - e_1 = s, \quad e_2 = p_2^2 - 2xp_3 + 2p_4, \quad e_3 = p_3^2 - 2p_2p_4, \quad e_4 = p_4^2.$$

Consequently $t + 8p_4 = 8xp_3$ and $p_3^2 = e_3 + sp_4$. Squaring and substituting yields

$$t^2 + 64(e_4 - x^2 e_3) + 16p_4(t - 4x^2 s) = 0.$$

Thus $C + 16p_4 L = 0$; eliminating p_4 proves that (5) vanishes at $x = \sum r_i$. The calculation applies to all sixteen sign choices. Those roots are distinct over the generic rational-function field. Both expressions are monic of degree sixteen in x, so they are equal. Polynomial specialization covers zero radicands and coincident roots; no division by a specialized quantity occurs.

The four elementary symmetric expressions cost twelve operations. The remaining schedule costs $1 + 3 + 5 + 3 + 5 = 17$, giving 29. □

2.2 The preferred 28-operation schedule

During review, a retained source proof was found to contain a 28-operation schedule for the same norm. It is explicitly credited and pinned in the packet. It gives the preferred final counts, one operation below the symmetric variants:

$$\begin{aligned} \sigma_1 &= a + b + c, & \sigma_2 &= ab + (a + b)c, & \sigma_3 &= abc, \\ U &= z + d, & P &= U - \sigma_1, & E &= P^2 + 4(zd - \sigma_2), \\ G_{\mathrm{e}} &= E^2 + 16(zdP^2 - 4\sigma_3 U), & G_{\mathrm{o}} &= EP - 16\sigma_3, \\ N_4(z; a, b, c, d) &= G_{\mathrm{e}}^2 - 64zdG_{\mathrm{o}}^2. \end{aligned} \tag{6}$$

The cost is 28 with shared intermediates. Pair the two choices of the fourth root in the octic three-root norm to derive (6), or compare all coefficients with (3). The delivered checker uses the latter independent method, obtaining all 495 coefficients.

The main computational improvement is to clear a rational radicand *before* applying either compact norm schedule:

$$\boxed{\delta^8 F_{16}(Z; A, B, E, C_{\mathrm{n}}/\delta) = N_4(\delta Z^2; \delta A, \delta B, \delta E, C_{\mathrm{n}}).} \tag{7}$$

The left side means its cleared polynomial. Homogeneity proves equality for nonzero δ, and polynomial identity extends it to $\delta = 0$. No actual division is required in the evaluator.

If $A = U^2 D_f$, $B = U^2 D_t$, $E = U^2 A_g$, and $C_{\mathrm{n}} = U^2 C_y$, form $D = \delta U^2$ once and evaluate

$$N_4(\delta Z^2; DD_f, DD_t, DA_g, U^2 C_y). \tag{8}$$

The shared scale saves two additional multiplications. With (6), the whole thirteen-integer evaluator drops from 135 to 126 operations. The alternative symmetric evaluator, with the elementary identity (5), has 127 operations and is also retained.

2.3 A complete thirteen-integer equation

The variables are

$$j, C, z, h, r, v, w, T, c, n, t, p_2, p_3 \in \mathbb{Z}. \tag{9}$$

The six fixed coefficients $\beta, K_0, M_C, M_F, A_{\mathrm{cone}}, B_{\mathrm{cone}}$ are the entire pinned mixed-parity tuple in `integer/coefficients/`. In particular,

$$\beta = 2^{1064921} - 1, \qquad A_{\mathrm{cone}} = 2^{1113}, \qquad B_{\mathrm{cone}} = 2^{666104}.$$

The other three numerals are supplied as complete, authenticated little-endian byte strings. These are fixed coefficients, not quantified inputs. This tuple differs from the BFS tuple of Appendix A. The supplied M_F already includes its single shift by β.

Define

$$\begin{aligned}
&J = 6j + 1, \quad q = \beta J + 1, \quad W = 12w + 4, \quad X = qW,\\
&Y = qv, \quad E_0 = XY, \quad a = E_0 + Y, \quad H = 4a + 3,\\
&\Delta = a^2 + H, \qquad F = (K_0 + W)C - z(q-1),\\
&R = [q(q-F) - C](q^2-1) + (M_C + qM_F)J,\\
&k = R + 1 + hE_0, \quad \eta = c - Yk, \quad \zeta = k - \eta,\\
&i = 2r + 1, \quad I = ic^2, \quad D_f = 1 + \Delta I^2, \quad Q = \Delta^2 I^2,\\
&\delta = Q - 1, \quad V = D_f(cT - R) - c, \quad C_y = QV^2 - 1,\\
&\ell = XY^2, \quad L_0 = \ell k, \quad D_t = 1 + L_0(L_0 + k).
\end{aligned} \tag{10}$$

The remaining expressions are

$$\begin{aligned}
G &= (A_{\text{cone}}F)(B_{\text{cone}}C) - (q - A_{\text{cone}}F - B_{\text{cone}}C)^2,\\
\Pi &= G\eta\zeta - (t^2 - G - \eta\zeta + 1)^2,\\
A_g &= \Pi - 1 - p_2^2 - p_3(p_3 + 1),\\
M &= 2q^2, \quad U = MH, \quad K = X(X + 2ac) - 1,\\
Z &= U(n + V) + MK + H(v - q^2), \qquad D = \delta U^2.
\end{aligned} \tag{11}$$

The equation is (8) equal to zero, with the explicit norm (6). It is an ordinary polynomial in precisely the thirteen variables (9). Identity (7) proves equality with the entire parent polynomial. Its exact degree is 536; for example the attained-degree coefficient is

$$[j^{240}w^{80}v^{96}h^{16}r^{32}c^{64}z^{8}]P = 2^{488}3^{320}\beta^{240} > 0. \tag{12}$$

The exponents sum to 536. Universal polynomial equality is why this coefficient, the degree, and every integer zero transfer.

2.4 Twelve-integer companions

The two companions use $j, C, z, h, r, v, w, T, c, n, t, s \in \mathbb{Z}$ and the same six coefficients. Their complete prefixes are unchanged from the pinned scalar-carrier sources. In the source notation, their new tails are

$$N_4(\delta Z^2; 9\delta S^2 D_f,\ 81\delta S^2 D_t,\ 729\delta S^2 D_g,\ S^2 C_y) \tag{13}$$

and

$$N_4(Z^2; 9S^2 D_f,\ 81S^2 D_t,\ 729S^2 D_g,\ S^2\delta C_y). \tag{14}$$

With (6) they have 138 operations at degree 1,200 and 137 at degree 1,456. The symmetric schedule (5) costs one additional operation in each case. The complete definitions of S, Z, D_g and every other front abbreviation are literal in the delivered JSON circuits and copied source proofs. These are exactly the original polynomials, with smaller evaluators.

3 The new ten-positive-variable equation

The only unknowns are

$$J, F, \alpha, i, T, w, Y, n, \eta, \zeta \in \mathbb{Z}_{>0}. \tag{15}$$

Use the four BFS coefficients β, K_0, M_C, M_F defined entirely in Appendix A. All other symbols below are polynomial abbreviations.

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq, \\ k &= \eta + \zeta, & c &= kY + \eta, & \mu &= q^2 + 1, \\ t &= \mu Y, & A &= t(X+1) + 1, & \Delta &= A^2 - 1, \\ H &= 4A - 5, & v_0 &= Xt^2, & I &= ic^2. \end{aligned} \tag{16}$$

Set

$$\begin{aligned} D_\tau &= 1 + v_0(v_0+1)k^2, & D_f &= 1 + \Delta I^2, \\ Q &= \Delta^2 I^2, & \delta &= Q - 1, \\ R &= [(q-1)(q-F) + \alpha](q^2-1) + (M_C + qM_F)J, & & \\ e &= k - R - 1, & B_t &= (K_0 + w)C - F, \\ b &= c(T+\alpha) - R, & V &= D_f b - c, \\ D_{\mathrm{n}} &= QV^2 - 1, & K_{\mathrm{pos}} &= X(3c - 2X) + 2, \\ W &= 1 + (q-1)b, & P_0 &= q^3(q-1), \qquad M = P_0\mu^2 H. \end{aligned} \tag{17}$$

Form

$$\begin{aligned} Z = P_0\mu^2[H(n+V) + K_{\mathrm{pos}}] \\ + H[\mu^2((q-1)Y - q^3 W B_t) - P_0 e], \end{aligned} \tag{18}$$

$$S = \delta Z^2, \quad B_\tau = \delta M^2 D_\tau, \quad B_f = \delta M^2 D_f, \quad B_y = M^2 D_{\mathrm{n}}, \tag{19}$$

$$E = (S + B_y - B_\tau - B_f)^2 - 4(SB_y + B_\tau B_f). \tag{20}$$

The single equation is

$$\boxed{P_{96} = E^2 - 64 S B_y B_\tau B_f = 0.} \tag{21}$$

It has exact degree 360 and a 96-operation evaluator. There is no square root, division, variable exponent, hidden coefficient, or additional quantified variable in (16)–(21). The circuit evaluates D_τ as $(v_0 k)(v_0 k + k) + 1$ and shares $A - 1 = (\mu Y)(X+1)$.

Relative to the pinned 99/368 parent, two changes are made: the factor 9 on B_τ is removed, and the transport factor $1 + (q-1)(HI + b)$ is replaced by $1 + (q-1)b$. The first requires separation of three radicals; the second requires a stronger transport bound. Neither change is justified by an identity of the complete old and new polynomials.

The arithmetic argument is uniform for coherent tuples satisfying

$$\beta \text{ odd}, \quad \beta \geq 31, \quad K_0 > 0, \quad 0 < M_C < \beta < M_F < 2\beta. \tag{22}$$

These inequalities are not themselves a consistency calibration. All four actual constants must be the complete calibrated tuple, as they are in the appendix and the reconstructed coefficient files.

4 A square gap that removes the radical weight

The parent uses $9B_\tau$ in place of B_τ, so its signed radical relation contains 3τ. Deleting that weight requires more than checking the intended integral witnesses: a rational signed sum of irrational roots could otherwise introduce extra solutions. The following argument excludes every such cancellation before any source semantics is used.

Lemma 4.1 (A square gap)**.** *Let $x \geq 2$ and $t \geq 1$ be integers. Put*

$$\Delta = t(x+1)(t(x+1)+2), \qquad v = t^2x, \quad D = xt(x+1)(t(x+1)+2)(t^2x+1).$$

Then $\Delta v(v+1) = t^2D$ is not a rational square. In particular $\Delta I^2 = v(v+1)k^2$ has no solution with nonzero integers I, k.

Proof. Set $m = \lfloor x/2 \rfloor$ and $h = t^2x(x+1) + tx + m$. For $x = 2m$, direct expansion gives

$$D - h^2 = m[2t^2 + 4(m+1)t - m] > 0, \tag{23}$$

$$(h+1)^2 - D = 2m(4m+1)t^2 - 4m^2t + (m+1)^2 > 0. \tag{24}$$

Here $m \geq 1$. The first is positive already after replacing t by 1. In the second, $t^2 \geq t$ gives the positive lower bound $(4m^2+2m)t + (m+1)^2$. For $x = 2m+1$, where $m \geq 1$, the exact identities are

$$D - h^2 = t^2x(x+2) + 2tx(m+2) - m^2 > 0, \tag{25}$$

$$(h+1)^2 - D = [tx - (m+1)]^2 > 0. \tag{26}$$

The first is positive at $t = 1$ and increases thereafter. In the second, $tx \geq 2m+1 > m+1$. Thus $h^2 < D < (h+1)^2$ in both parity classes. An integer that is a rational square is an integer square, by reduction of its rational square root to lowest terms. Finally the prohibited equation would imply $D = (\Delta I/(tk))^2$. The excluded point $x = t = 1$ gives $D = 16$; the domain is essential. □

Lemma 4.2 (Rationality of three separated radicals)**.** *Let a, b, c be positive, pairwise distinct rational numbers with*

$$(a+b-c)^2 \neq 4ab. \tag{27}$$

If $u^2 = a$, $v^2 = b$, $w^2 = c$ and $u+v+w$ is rational, then u, v, w are rational.

Proof. Write $s = u+v+w$. If $s = 0$, squaring $w = -u-v$ twice contradicts (27), so $s \neq 0$. For $K_a = s^2 + a - b - c$, squaring $2vw = K_a - 2su$ gives

$$4sK_au = K_a^2 + 4s^2a - 4bc. \tag{28}$$

At least one of K_a, K_b, K_c is nonzero, since otherwise subtracting them would give $a = b = c$. The corresponding identity makes one root rational. The sum of the other two is rational and nonzero, since their squares differ. If that sum is $r = v + w$, then $v = (r^2 + b - c)/(2r)$ and $w = r - v$. □

Theorem 4.3 (Unweighted separation)**.** *Use Δ, v from Lemma 4.1. Suppose an integer $\mu > 3$ divides t, integers I, k satisfy $I \geq k \geq 1$, and V is a nonzero integer. Put*

$$d_\tau = 1 + v(v+1)k^2, \quad d_f = 1 + \Delta I^2, \quad Q = \Delta^2 I^2, \quad d_y = \frac{QV^2 - 1}{Q - 1}. \tag{29}$$

Every rational signed sum of $\sqrt{d_\tau}, \sqrt{d_f}, \sqrt{d_y}$ has all three radicals rational.

Proof. We have $\Delta > v + 1 > 1$, so $Q > 1$ and all three radicands are positive. Lemma 4.1 excludes $d_\tau = d_f$. If $d_\tau = d_y$, then $Q(d_\tau - V^2) = d_\tau - 1$, but its right side is strictly between 0 and Q, since $v(v+1)k^2 < \Delta^2 I^2$. If $d_f = d_y$, rearrangement gives $d_f - V^2 = 1/\Delta$, contradicting integrality. Thus they are pairwise distinct.

For the remaining triangle obstruction, put $\delta = Q - 1$ and $D_{\mathrm{n}} = QV^2 - 1$. Equality in (27) would imply

$$[\delta(d_\tau + d_f) - D_{\mathrm{n}}]^2 - 4\delta^2 d_\tau d_f = 0. \tag{30}$$

Modulo μ we have $\Delta = v = Q = 0$, $\delta = D_{\mathrm{n}} = -1$, and $d_\tau = d_f = 1$. The left side is -3 modulo μ, impossible for $\mu > 3$. Lemma 4.2 applies, with arbitrary signs on the roots. □

4.1 Application on the full positive input domain

In (16), $X \geq 32$, $t = \mu Y$, $\mu > 3$, $c = kY + \eta > k$ and $I = ic^2 \geq c^2 > k$. Also $D_f > 2c$ and b is integral. If $b \leq 0$, then $V = D_f b - c \leq -c$; if $b \geq 1$, then $V \geq D_f - c > c$. Thus $V \neq 0$ before the polynomial equation is used. Every hypothesis of Theorem 4.3 is unconditional on the supplied positive domain.

The octic identity

$$F_8(s; a, b, c) = [(s + c - a - b)^2 - 4(sc + ab)]^2 - 64scab \tag{31}$$

factors, with $s = x^2$, into the eight signed linear radical factors. Since $M, \delta > 0$, equation (21) implies

$$\frac{Z}{M} = \epsilon_1 \tau + \epsilon_2 f + \epsilon_3 y, \qquad \tau^2 = D_\tau, \quad f^2 = D_f, \quad y^2 = D_{\mathrm{n}}/\delta, \tag{32}$$

with positive roots. Theorem 4.3 makes all three roots rational. The first two are integers because their squares are integers; the retained adjacent-Pell descent lemma makes the third integral under the unchanged $Q = \Delta^2 I^2$ and integer-V hypotheses [1].

5 The auxiliary Pell index supplies its own transport bound

Removing the weight alone gives 98 operations at degree 368. We now show that the already encoded auxiliary Pell solution grows enough to remove the separate HI term in the transport majorant. This is the additional reduction to (21).

For $A \geq 2$, write

$$\chi_A(m) + \psi_A(m)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^m.$$

The unchanged strong/auxiliary folding argument, after the integral roots are recovered, gives $c = \psi_A(p)$ for an odd positive p. The parity does not require the later equality $p = R$: the auxiliary index is odd, and, with m the strong Pell index, both reduction modulo $2m$ and reflection from an index u to $2m - u$ preserve oddness. Since $c \geq 3$, $p = 1$ is excluded, hence

$$c \geq \psi_A(3) = 4A^2 - 1 = 4\Delta + 3. \tag{33}$$

Before transport, the retained first-index growth and mask estimate give

$$c > 8X\mu^2 Y^2, \qquad R < c\alpha/4. \tag{34}$$

These bounds do not use $C > 0$, the main index equality, or the sign of B_t. In particular

$$b = c(T + \alpha) - R > c(T + 3\alpha/4) > c. \tag{35}$$

Lemma 5.1 (Auxiliary index at least five)**.** *Under the recovered kernel equations and (33)–(35),*

$$b > 8Q > 2HI. \tag{36}$$

Proof. Set $S_0 = \Delta I$, so $Q = S_0^2$. The auxiliary equation is $S_0^2V^2 - (S_0^2 - 1)y^2 = 1$. Here $V > 0$. Pell classification gives $S_0V = \chi_{S_0}(\nu)$ at a positive index ν. Even indices have $\chi_{S_0}(\nu) \equiv \pm 1 \pmod{S_0}$, whereas S_0 divides S_0V, so ν is odd.

By (33)–(35),

$$V = D_f b - c > \Delta I^2 c > 4\Delta^2 I^2 = 4Q.$$

The possibilities $\nu = 1$ and $\nu = 3$ would give respectively $V = 1$ and $V = 4Q - 3$, and are impossible. Therefore $\nu \geq 5$, and monotonicity gives

$$V \geq 16Q^2 - 20Q + 5 > 8Q^2. \tag{37}$$

The last inequality holds for $Q \geq 3$, far below the actual lower bound. Because $D_f = \Delta I^2 + 1 < Q$, $b = (V + c)/D_f > 8Q$. Finally $\Delta - H = (A - 2)^2 > 0$ gives $8Q > 2HI$. □

Lemma 5.2 (Quarter bounds for the two error terms)**.** *With $u = y - V$, the recovered positive roots satisfy*

$$0 < u < b/4, \qquad e/\mu^2 < b/4. \tag{38}$$

Proof. The auxiliary equation gives

$$u = \frac{V^2 - 1}{(Q - 1)(y + V)} < \frac{V}{2(Q - 1)} < \frac{D_f b}{2(Q - 1)}.$$

Here $y > V > 1$, and $Q - 1 - 2D_f = \Delta(\Delta - 2)I^2 - 3 > 0$. Thus $u < b/4$. Also $b - e = c(T + \alpha) - k + 1 > 0$, so $e < b$. Since $b > 0$ and $\mu^2 > 4$, $e/\mu^2 < b/4$. This upper bound also holds when e is negative; no premature divisibility or positivity of e is assumed. □

5.1 Soundness of the shorter carrier

The retained bound $3\tau + f < HI$ implies $\tau + f < HI < b/2$. The normalized center from (18) is

$$\frac{Z}{M} = n + V + \frac{K_{\mathrm{pos}}}{H} + \frac{Y}{q^3} - \frac{WB_t}{q - 1} - \frac{e}{\mu^2}. \tag{39}$$

Suppose $B_t \leq -1$. With the new $W = 1 + (q - 1)b$, $WB_t/(q - 1) < -b$. The parent growth estimate already gives $K_{\mathrm{pos}} > 0$. Using (32), (36) and (38),

$$\begin{aligned} n &= \epsilon_1\tau + \epsilon_2 f + (\epsilon_3 y - V) - K_{\mathrm{pos}}/H - Y/q^3 + WB_t/(q - 1) + e/\mu^2 \\ &< b/2 + b/4 + b/4 - b = 0, \end{aligned}$$

contrary to $n > 0$. Thus $B_t \geq 0$ and hence $C > 0$.

Everything after this transport step is unchanged. In the original order the proof recovers $3q < R < q^4 - 2$, $p = R$, parity, and then $\mu^2 \mid e$. The first index is $(R + 1)/2$. The ratio estimate and whole-product projection force $X = 2^R$; no noninvertible factor is cancelled modulo H. The remaining transport and half-binomial conditions reconstruct the same unscaled native source and a finite syntactic refutation. The new W retains residue 1 modulo $q - 1$.

5.2 Completeness

Use the parent's fresh canonical zero-offset witnesses from a finite refutation. Retain every coordinate except n. At these witnesses

$$s_0 = Y/q^3 \geq 1, \quad t_0 = B_t/(q-1) \geq 0, \quad \rho = (k-R-1)/\mu^2 \geq 1, \quad u = y - V > 0, \quad \kappa = K_{\mathrm{pos}}/H > 0$$

are integers. With the new W, choose

$$n = Wt_0 + \rho + \tau + f + u - s_0 - \kappa. \tag{40}$$

The parent gives $\tau > k > Y \geq s_0$ and $f > \kappa$. Also $W > 0$ because $b > c > 0$, so $n > 0$. Equation (39) becomes $Z/M = \tau + f + y$, and a norm factor vanishes. This is canonical reconstruction, not an asserted positive pointwise inverse from every parent tuple.

Corollary 5.3. *At the coherent coefficients of Appendix A, P_{96} has a positive integer zero exactly when the pinned ZFC calculus has a finite refutation. Therefore ordinary mathematics formalizable in ZFC gives (1) with $D = \mathbb{Z}_{>0}$ and $n = 10$.*

6 Exact degrees and the direct-base companion

The degree must be calculated for the entire norm. Following only the centered difference can miss a different leading term. Write

$$s = \delta Z^2, \quad a = \delta M^2 D_\tau, \quad b_* = \delta M^2 D_f, \quad c_* = M^2 D_{\mathrm{n}}, \quad d = s - c_*.$$

Then

$$E = d^2 - 2(s + c_*)(a + b_*) + (a - b_*)^2. \tag{41}$$

For the new scaled carrier the relevant degrees are

$$\deg a = 74, \quad \deg b_* = 76, \quad \deg s = \deg c_* = 104, \quad \deg d = 86.$$

The term d^2 has degree 172, but $-2(s + c_*)b_*$ has degree 180 and dominates it. Since s and c_* have the same leading component,

$$E_0 = -4(c_*)_0(b_*)_0, \qquad (P_{96})_0 = 16(c_*)_0^2(b_*)_0^2. \tag{42}$$

The subtracted norm term has degree 358, below 360. Thus the exact degree is **360**. Its full leading component is

$$2^{20} q_0^{184} w^{36} i^{20} Y^{76} (\eta + \zeta)^{40} (q_0 - F)^4, \qquad q_0 = \beta J. \tag{43}$$

For example,

$$[J^{184} w^{36} i^{20} Y^{76} \eta^{40} F^4] P_{96} = 2^{20} \beta^{184} > 0. \tag{44}$$

6.1 The direct-base equation of degree 332

Replace w by a positive input x and set $X = q + x$. Keep the same zero-offset main parameter and radicands. Use

$$B = (qK_0 + X)C - qF, \quad W = 1 + q(q-1)b, \quad P_0 = q^4(q-1), \quad M = P_0 \mu^2 H,$$

$$\begin{aligned} Z = P_0\mu^2[H(n+V)+K_{\mathrm{pos}}] \\ + H[\mu^2(q(q-1)Y - q^3WB - (q-1)X) - P_0 e]. \end{aligned} \tag{45}$$

Evaluate the same unweighted norm (19)–(21). The square gap and auxiliary index argument use $X \geq 32$, not $q \mid X$, so apply before the direct carrier recovers that divisibility. The retained inequality $3\tau + f + X/q^4 < HI < b/2$ supplies the corresponding transport estimate.

For canonical completeness the direct parent has the exact map

$$x = X - q, \qquad n_{\mathrm{direct}} = n_{\mathrm{scaled}} + (q-1)^2 b t_0 + X/q^4 > 0. \tag{46}$$

Its numerator and denominator scale by q and the norm output by q^8. The late-divisibility proof and original source reconstruction remain valid [2].

The direct evaluator has 102 operations. Its norm degrees are $(68, 70, 96, 96, 80)$, so the degree is 332. The leading component is

$$2^{20} q_0^{156} X_0^{36} i^{20} Y^{76} (\eta + \zeta)^{40} (q_0 - F)^4, \qquad X_0 = q_0 + x. \tag{47}$$

The coefficient of $J^{156} x^{36} i^{20} Y^{76} \eta^{40} F^4$ is $2^{20}\beta^{156} > 0$. This companion trades six more operations for 28 fewer degrees.

6.2 Retained intermediate equations

The packet retains the weight-only reductions, before shortening the transport carrier: 98 operations at degree 368 and 104 at degree 340. They isolate the new radical-separation theorem. The final 96/360 and 102/332 endpoints improve both degree and operations.

For the weight-only step the exact identity

$$P_9 - P_1 = 16a(5a - s - b_* - c_*)(E_9 + E_1) - 512 s c_* a b_* \tag{48}$$

has nonzero differences of exact degrees 362 and 334 respectively, where $E_\lambda = d^2 - 2(s + c_*)(\lambda a + b_*) + (\lambda a - b_*)^2$. Even this first step changes the polynomial, unlike the integer norm rewrites.

7 Merging sign certificates in the direct quartic

The direct quartic uses a different source representation. Its arithmetic body includes quotient/remainder rows, with separate nonnegativity certificates for a remainder and its complementary slack. A shared positive modulus lets two such certificates be replaced by one.

Lemma 7.1 (Shared remainder bounds)**.** *For integers r, s, B with $r + s + 1 = B$ and $B \geq 1$,*

$$(r \geq 0 \ \wedge \ s \geq 0) \quad \longleftrightarrow \quad rs \geq 0. \tag{49}$$

Proof. The forward implication is immediate. Conversely, nonnegative product excludes opposite nonzero signs. Both values cannot be negative because their sum is $B - 1 \geq 0$. If either is zero, the other equals $B - 1 \geq 0$. □

For integer variables, nonnegativity has the representation

$$n = a^2 + b^2 + d(d+1). \tag{50}$$

The right side is nonnegative. Conversely, $4n+1$ for $n \geq 0$ is not of the excluded form $4^u(8v+7)$, so the three-square theorem represents it by three squares. Exactly one coordinate is odd, by reduction modulo four. Relabeling gives $4n+1 = (2a)^2 + (2b)^2 + (2d+1)^2$, proving (50).

Two linear nonnegative expressions normally require six private integer witnesses. Their product instead uses three, with the quadratic row $rs - a^2 - b^2 - d^2 - d = 0$. The modulus sign is supplied by a retained anchor pair with the same sum; no source interpretation is needed.

7.1 The six literal pairs

The indices below refer to the original arithmetic body, before projection. Each anchor remains separately guarded, and its exact source rows prove that the shared modulus is at least one.

Remainder	Slack	Modulus	Retained anchor pair
562	563	547	549, 550
585	586	574	576, 577
593	594	574	576, 577
601	602	574	576, 577
619	620	608	610, 611
637	638	626	628, 629

Table 2: Six applications of (49). The coefficient certificate checks each $r + s + 1 - B = 0$ identity by summing actual source rows.

For example, raw rows 553–555 are

$$-v_{585} - v_{586} + v_{589} = 0, \quad -1 - v_{589} + v_{590} = 0, \quad v_{574} - v_{590} = 0.$$

Their sum is $v_{574} - v_{585} - v_{586} - 1 = 0$. Three other unchanged rows give $v_{574} - v_{576} - v_{577} - 1 = 0$. The retained guards on v_{576} and v_{577} establish $v_{574} \geq 1$; the new guard on $v_{585}v_{586}$ therefore restores both omitted signs.

The six replacements retain 63 old predicates and add six product predicates, giving 69 instead of 75. Every arithmetic coordinate and body equation is unchanged. A new zero restores the twelve omitted nonnegative values by (49), and fresh witnesses from (50) give the complete old system. An old zero makes each product nonnegative, so fresh product witnesses give the new system. Thus their projections onto the arithmetic body coordinates are identical.

7.2 One strict lower bound replaces two more signs

There is a further exact implication in the same literal body:

$$v_7 = v_6^2,\ v_8 = v_7^2,\ \ldots,\ v_{28} = v_{27}^2, \qquad v_{28} = v_3 + 1. \tag{51}$$

There are 22 squaring rows, followed by the two original rows establishing the last equality. On these integer equations,

$$(v_3 \geq 0 \ \wedge\ v_6 \geq 0) \quad \longleftrightarrow \quad v_6 \geq 1. \tag{52}$$

Indeed $v_6 \geq 1$ propagates through the squares to $v_{28} \geq 1$, giving $v_3 \geq 0$. Conversely $v_6 = 0$ would force $v_3 = -1$, contradicting the old guard; the old $v_6 \geq 0$ then implies $v_6 \geq 1$. This covers the boundary $v_6 = 1, v_3 = 0$.

Replace the two old triples by the single guard

$$v_6 - 1 - a^2 - b^2 - d(d+1) = 0. \tag{53}$$

The source now has 68 predicates. A new zero first restores the 69-predicate system by (51)–(52); the shared remainder proof then restores all 75 original predicates. Only after those restorations is the C222 source interpretation applied. Completeness uses fresh triples in the reverse order. No omitted sign is justified by a semantic interpretation that itself requires that sign.

7.3 The complete 589-variable quartic

The unprojected system has $1,259 + 3 \cdot 68 = 1,463$ integer coordinates and $1,177 + 68 = 1,245$ quadratic rows. The principal monic projection removes 874 coordinates and rows, leaving 589 coordinates and 371 quadratic residuals. Each deleted coordinate has a unique integral restoration, checked by substituting in every remaining equation. There is no rational relaxation or unrecorded free variable.

Let the delivered residuals be $f_0, \ldots, f_{370}$. The equation is

$$\boxed{Q_{589}(x_0, \ldots, x_{588}) = \sum_{j=0}^{370} f_j(x_0, \ldots, x_{588})^2 = 0, \qquad x_i \in \mathbb{Z}.} \tag{54}$$

Every coefficient of every residual is explicitly supplied in `quartic/fused68/support8/equation.json`. The fully collected polynomial is in the adjacent `polynomial.json`, and the plain-text residual list gives the same finite specification without a JSON reader.

Over $\mathbb{Z}$, (54) is equivalent to all residuals being zero. Projection followed by guard restoration recovers the entire old 75-predicate source. Applying its ordinary C222/refutation calibration therefore gives (1) with $D = \mathbb{Z}$ and $n = 589$.

Profile	Variables	Rows	Monomials	Height
Matched old source	604	372	9,676	2,445,790,974,704
Six pair replacements	592	372	8,167	2,445,790,974,704
Final principal	**589**	371	8,156	2,445,790,974,704
Final, quadratic first	595	377	7,595	6,956,012
Final, smaller coefficients	600	382	7,075	964,213

Table 3: Complete unrestricted-integer quartics, all of exact degree four. The 600-variable alternative has every coefficient below one million.

The guard count alone would predict an eighteen-variable saving from the six pairs. The actual projection saves twelve, because the products keep six additional source coordinates alive. The strict bound then saves three more variables. All reported counts are from complete equations and checked inverses, rather than estimates.

7.4 The direct logical endpoint

The unchanged arithmetic body directly encodes the numerical C222 calculus. The named logical endpoint is the composition of

`KEndpoint.c222_iff_zf_refutation`

and

`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

The first identifies C222 derivation of its falsehood code with finite ZF refutation; the second transports refutability between ZF and ZFC. Their exact immediate source files are included and pinned, but their transitive Lean imports were not rebuilt here. The ordinary arithmetic bridge and later guard reductions are explicitly identified in `quartic/CALIBRATION_DEPENDENCIES.md`. The equation uses neither K215 simulation nor Jones' universal polynomial.

8 Verification, reproducibility and proof boundaries

Exact algebra. The norm checker multiplies all sixteen conjugate factors and compares all 495 coefficients with the new schedules. For each of six delivered integer variants, it proves equality of the complete old and new tails over a free polynomial ring and authenticates every unchanged prefix. The positive checker verifies the parity identities, octic norm, complete register inventories, exact leading polynomials and degree claims. The quartic checker verifies source-row identities, all monic integral restorations, and every collected coefficient of the entire sum of squares.

Finite specification and provenance. The files contain every coefficient and every operation. The BFS reconstructor decodes the entire 436-character seed, constructs all 430 actions and 943 tiles, and reconstructs all four full integers. Each raw payload matches the pinned complete-page value, rather than only a modular residue or selected bits. The integer packet includes all six authentic coefficient payloads, and the direct quartic has literal coefficients with no parameter ports.

Unbounded mathematics. The square gap, radical rationality, index bounds, transport contradiction, guard reconstruction and consistency calibration concern arbitrarily large witnesses. The paper proofs supply those implications. Modular controls are only supplementary transcription checks. Neither a finite search finding no zeros nor large numerical experiments would prove consistency.

8.1 Claims that are not being made

This packet supplies no new complete Lean or Rocq certificate, no encoded object-language derivation of the entire calibration, and no proof of Con(ZFC) itself. The source pins and retained licenses identify the inherited dependencies. We did not rerun every historical native compiler or earlier proof-assistant build in the repository.

The operation count is not a bit-complexity bound. The fixed native coefficients are large, and legitimate witnesses may be much larger. The direct quartic has smaller literal coefficients but many more variables. No result is advertised as uniformly best in all those measures.

8.2 Reproduction

Run from the extracted packet:

```
python3 verify_all.py
```

The exact checkers use the Python standard library. They check the supplied finite artifacts without network access; optional builders are included too.

The report is rebuilt with:

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

Recorded receipts and a checksum manifest accompany the files.

9 What this round establishes

The direct integer result saves fifteen actual variables and more than fifteen percent of the expanded support at unchanged degree and coefficient height. It uses evidence already present in the source: a common positive remainder bound and a short squaring chain.

The compact positive result uses two genuine proof improvements. The square-gap lemma replaces an asymmetric radical weight with a uniform rationality argument. The auxiliary index bound then shows that the existing witness already supplies enough growth for transport, eliminating two additional arithmetic operations. The final scaled and direct equations improve both degree and operation count over their respective parents.

The integer norm changes are algebraically simpler to certify. Because the complete polynomials agree identically, there is no new signed-domain soundness argument or exceptional specialization to resolve. The packet retains both the symmetric derivation and the one-operation cheaper schedule found in the pinned source.

Two tempting reductions were not promoted to theorems. Replacing $\mu = q^2 + 1$ by $q + 1$ destroys the first-index pinning inequality used with $R < q^4$. Replacing the trace's additive positivity certificate by a Pell guard can leave a zero-ordinate branch. No delivered endpoint relies on either unproved replacement.

A substantially smaller variable count remains open. The present constructions are concrete improvements with reproducible equations and proofs, not evidence that a global minimum has been reached.

A Complete fixed coefficients for the positive equations

This is the finite BFS compact44 recipe used by the pinned complete equation page [6]. It defines all four coefficients in (16)–(21) without an external coefficient oracle. Every index and exponent is fixed or bound in a finite sum. The script `reconstruct_constants.py` executes this recipe independently and compares all four entire raw byte strings with the pinned values. The letter K in this inherited recipe denotes K_0 in the main text.

The four fixed integers

Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{t \in \mathcal{T}} v(t) x^{\varrho(t)} \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} = {} & \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ & (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ & (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 fixed tiles and their ranks

Use the actions u_j, v_j, d_j decoded below. Let

$$h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\},$$
$$\mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\},$$
$$r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.$$

An interval ending at -1 is empty. The following union is $\mathcal{T}$; the index ranges in each row are fixed:

range	tile (w, n, e, s)
$j = 2q + b < 430$	$((1 - d_j) l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$b = 0, 1$	$(0, h(215, b), 0, h(215, b)),\ (0, b, 0, b)$
$q \in \mathcal{R},\ b = 0, 1$	$(r_q, b, 0, h(q, b))$
$q \in \mathcal{L},\ b = 0, 1$	$(0, b, l_q, h(q, b))$
$t \in \{0, 1, 432, 433\}$	$(0, t, 0, 438)$
	$(2, 434, 2, 0),\ (2, 435, 3, 1)$
	$(3, 436, 4, h(u_0, 0)),\ (4, 436, 4, 0),\ (5, 0, 0, 0)$
$t = 434, 435, 436$	$(1, 438, 1, t)$
$a \in \{0, 1, 4\},\ b = 0, 1, 2$	$(a, 437, b, 437)$

Put $s_* = (2, 435, 3, 1)$, $f_* = (5, 0, 0, 0)$, and $\mathcal{T}' = \mathcal{T} \setminus \{s_*, f_*\}$. For $t = (w, n, e, s)$ set

$$\kappa(t) = 2^{30}w + 2^{20}n + 2^{10}e + s,$$
$$v(t) = w + 2^8 n + 2^{17}e + 2^{25}s + 127\,2^{34},$$
$$\varrho(s_*) = 0, \quad \varrho(f_*) = 1,$$
$$\varrho(t) = 2 + \sum_{z \in \mathcal{T}'} \left\lfloor \frac{2^{40} + \kappa(t) - \kappa(z) - 1}{2^{40}} \right\rfloor \quad (t \in \mathcal{T}').$$

A closed decoder for the 430 actions

Put $D_b(A, k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \lfloor \mathcal{A}/2^{636} \rfloor \bmod 2^{1675},$$
$$E_0 = \lfloor \mathcal{A}/2^{2311} \rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The fixed seed integer

The alphabet, in order of digit values $0, \ldots, 61$, is `0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`.

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTteOeUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

B Delivered file organization

Path	Contents
`report.tex`, `report.pdf`	The report, with its companion LaTeX input files.
`positive/`	Final 96/360 and 102/332 circuits; weight-only checkpoints; mathematical proofs, exact checks, original coefficients and pinned source proofs.
`positive/coefficients_bfs/`	The four completely reconstructed coefficients used by the main displayed equation, with a binding for the BFS circuit variants.
`integer/`	Preferred 126-, 138-, and 137-operation integer circuits, symmetric alternatives, original parents, all six coefficients and exact norm checks.
`quartic/`	All quadratic residuals, complete quartic coefficients, source predicates, projection certificates, builders, exact checkers and matched parent profiles.
`reconstruct_constants.py`	Finite reconstruction from the printed seed, actions and tiles.
`verify_all.py`	Runs the selected exact checks and records their outcomes.
`PROVENANCE.json`	Frozen repository snapshot, source references and claim boundaries.
`SHA256SUMS.txt`	Hashes of package files, excluding this checksum file itself.

References

[1] ProveIt contributors. *Ten-positive zero-offset equation*. https://github.com/VladimirReshetnikov/ProveIt/tree/84e16e43092614bae262ca59651cb9d77e31a255/Computability/DiophantineConsistency/candidates/ten-positive-zero-offset.

[2] ProveIt contributors. *Direct positive base with the zero-offset main parameter*. https://github.com/VladimirReshetnikov/ProveIt/tree/84e16e43092614bae262ca59651cb9d77e31a255/Computability/DiophantineConsistency/candidates/ten-positive-direct-zero-offset.

[3] ProveIt contributors. *Thirteen-integer quadratic trace equation*. https://github.com/VladimirReshetnikov/ProveIt/tree/84e16e43092614bae262ca59651cb9d77e31a255/Computability/DiophantineConsistency/candidates/integer13-quadratic-trace.

[4] ProveIt contributors. *Scalar carrier in twelve unrestricted integers*. https://github.com/VladimirReshetnikov/ProveIt/tree/84e16e43092614bae262ca59651cb9d77e31a255/Computability/DiophantineConsistency/candidates/scalar-carrier12.

[5] ProveIt contributors. *Complete 75-predicate direct quartic family*. https://github.com/VladimirReshetnikov/ProveIt/tree/84e16e43092614bae262ca59651cb9d77e31a255/Computability/DiophantineConsistency/research/factorial-radius-output-sign-profile.

[6] ProveIt contributors. *Complete ten-positive equation page and coefficients*. https://github.com/VladimirReshetnikov/ProveIt/tree/84e16e43092614bae262ca59651cb9d77e31a255/Computability/DiophantineConsistency/candidates/one-page-short10.

[7] Zhi-Wei Sun. *Ten unknowns for Hilbert's tenth problem over the integers*. arXiv:2609.01594, 2026. https://arxiv.org/abs/2609.01594.